VERA Virtual Environment for Reactor Applications

ORNL/SPR-2022/2530

# VERAIn Software Requirements, Test Plan, and Test Report



July 27, 2022

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ORNL/SPR-2022/2530




# VERAIn SOFTWARE REQUIREMENTS, TEST PLAN, AND TEST REPORT

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# VERAIn Software Requirements, Test Plan, and Test Report

## Revision Log

| Revision | Date | Affected Pages | Revision Description |
|---|---|---|---|
| 0 | 6/25/2022 | All | Implements new template and editorial updates for VERA 4.3 release. This version supersedes previous document version CASL-U-2019-1891-002. |

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# VERAIn Software Requirements, Test Plan, and Test Report

**Approvals:**

| | |
|---|---|
| *Erik Walker* | 08/09/2022 |
| **Erik Walker, VERAIO Product Software Manager** | **Date** |
| *Aaron Graham* | Aug 9, 2022 |
| **Aaron Graham, Independent Reviewer** | **Date** |

## EXECUTIVE SUMMARY

This document describes the software test plan for VERAIn and provides appendices for the software requirements and software test report. In this document, the test platform hardware and software are described. Appendix A provides a list of the tests run and their acceptability as a test report, and Appendix B provides the list of low-level software requirements as a requirements traceability matrix.

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# 1. TESTING PROCEDURE

## 1.1 COMPUTER PROGRAM TESTED

### 1.1.1 Program: VERAIn

Built under VERA Environment with following packages and versions:

**Listing 1. Software Versions Evaluated**

```
*** Base Git Repo: VERA
a896f8f [Thu Jul 7 13:20:22 2022 -0400] <bairdml@ornl.gov>
Adjusting checkout and submodule command fort production pipelines
*** Git Repo: TriBITS
9d0198a [Tue Apr 6 18:41:53 2021 +0000] <collinsbs@ornl.gov>
Merge branch 'timeout_issues' into 'master'
*** Git Repo: Trilinos
3b9c7e3 [Thu Jan 30 00:18:02 2020 -0500] <collinsbs@ornl.gov>
Merge pull request #1 from bartlettroscoe/casl-phi-6249-upgrade-tribits-trilinos
*** Git Repo: TeuchosWrappersExt
781aae8 [Fri May 10 09:51:36 2019 -0600] <rabartl@sandia.gov>
Add missing std include (PHI-5294)
*** Git Repo: Futility
2c86eec [Wed Jun 15 20:55:43 2022 +0000] <salkork@ornl.gov>
Fix to the requirements generator script
*** Git Repo: MAMBA
ad795e0 [Thu Sep 16 04:00:43 2021 -0400] <collinsbs@ornl.gov>
Merge branch 'master' of code-int.ornl.gov:mamba/MAMBA
*** Git Repo: VERAIO
2ceb8d2 [Wed Jul 13 11:26:28 2022 -0400] <ew4@ornl.gov>
Add tech reviewer changes to VERAIn User's Manual
*** Git Repo: DataTransferKit
19526ab [Fri Aug 18 09:14:16 2017 -0400] <dalg24@gmail.com>
Merge pull request #297 from naughtont3/tjn-fortran-off
*** Git Repo: COBRA-TF
69e3e14 [Mon Jul 11 12:32:30 2022 +0000] <salkork@ornl.gov>
Updates to the CTF documentation for the 4.3 release
*** Git Repo: VERAData
eb21c20 [Thu Jun 3 13:29:31 2021 +0000] <grahamam@ornl.gov>
Merge branch '4251_MPACT_v5.1m0_60n19g' into 'master'
*** Git Repo: VERAData/CEData
8252504 [Wed Jul 8 20:44:25 2020 -0400] <pandyatm@ornl.gov>
Merge branch 'update-hydrogen-pole-data' into 'master'
*** Git Repo: VERAOneWay
6af5147 [Tue Sep 14 13:46:59 2021 +0000] <collinsbs@ornl.gov>
Merge branch 'bison_post_states' into 'master'
*** Git Repo: SCALE
1a8cd5c [Tue Aug 17 12:12:19 2021 +0000] <pandyatm@ornl.gov>
Merge branch 'sync-scale' into 'master'
```

```
*** Git Repo: XSTools
08adb77 [Mon May 24 20:33:17 2021 +0000] <collinsbs@ornl.gov>
Merge branch '4874_XSTools_wrong_backXS_5x' into 'master'
*** Git Repo: MPACT
ad94f0c [Thu Jul 21 10:44:35 2022 -0400] <grahamam@ornl.gov>
Fix bad rebaseline of ge14_single_ARI
*** Git Repo: VeraShift
767cddc [Fri Aug 27 10:44:46 2021 -0400] <bairdml@ornl.gov>
Updating trigger for DownStream testing.
*** Git Repo: DakotaExt
afd354f [Tue Aug 1 19:36:55 2017 -0400] <rhoope@sandia.gov>
Fix tarball installation
*** Git Repo: DakotaExt/Dakota
7415238 [Thu Jun 15 12:40:49 2017 -0400] <rhoope@sandia.gov>
Triage static builds of Dakota 6.6
*** Git Repo: VUQDemos
b2d0175 [Tue Oct 8 14:13:19 2019 -0400] <salkork@ornl.gov>
Merge branch '1-correcting-hardcoded-python-paths-in-vuq-core' into 'master'
*** Git Repo: VERAView
28e0f8a [Thu Mar 31 16:53:16 2016 -0400] <leerw@ornl.gov>
Sync from code.ornl.gov for version 1.0
```

VERAIn has no dependencies on any other code package contained within VERA.

#### 1.1.2 System Software

These tests were performed on the Centos 7.9.2009 Linux 3.10.0-1160.66.1.el7.x86_64 operating system.

#### 1.1.3 Test Calibration

None.

#### 1.1.4 Computer Hardware Used for Testing

casl-vera01.ornl.gov

### 1.2 TEST EQUIPMENT CALIBRATION

None.

### 1.3 DATE OF TEST

Started: July 21, 2022 - 15:18:34 UTC
Total Test Time: 2 minutes, 30 seconds

### 1.4 DATA RECORDER

ctest-3.17.3 and CDash Version 2.5.0

### 1.5 SIMULATION MODELS USED

Because VERAIn is only an input parser, it contains no simulation models.

## 1.6 TEST PROBLEMS

A list of test problems and their requirements may be found in Appendix B of this document.

## 1.7 RESULTS AND APPLICABILITY

### Results

Results are given in Appendix A.

### Applicability

The test results from Appendix A satisfy the defined software requirements.

## 1.8 ACTION TAKEN IN CONNECTION WITH NOTED DEVIATIONS

There were no noted deviations. Therefore, no actions were taken.

### 1.8.1 Person Evaluating Results

Erik Walker

## 1.9 ACCEPTABILITY

The acceptance criterion is that all tests pass their individual criteria. As evidenced in column 4 of the table in Appendix A, all tests passed their criteria. The primary type of test performed, as shown in the VERAIn test matrix, was a regression test to compare a test output file to a version-controlled reference file deemed to be correct.

Regression test reference files are XML.gold files that have been generated and reviewed for correctness. Once the VERAIn parser is run on an input, the resulting XML file is compared to the corresponding XML.gold file using a straight diff. If the files are identical, then the test passes. If any differences exist in the files, then the test fails.

A few tests are designed to intentionally fail to capture common input mistakes and to ensure that the parser fails hard when these mistakes are encountered.

The test results were reviewed by the product support manager and the independent technical reviewer, who determined that the computer program is acceptable for integration with the VERA product suite. The test results demonstrated that the computer program(s) performed adequately and correctly. Testing covered the specified software requirements, the proper handling of abnormal conditions or events, and the interfaces with other components. Testing verified that the computer program did not introduce unintended consequences which degraded the overall software function. No discrepancies between expected and observed software functioning were identified.

## 2. APPLICABLE STANDARDS AND PROCEDURES

- NQA-1-2008 with NQA-1a-2009 Addenda [1]
- VERA-QA-004, VERA Software Configuration and Control
- VERA-QA-006, VERA Software Requirements, Design, and Testing

# 3. REQUIRED RECORDS

## 3.1 CDASH TEST RESULTS

See Appendix A.

## 3.2 REQUIREMENTS TRACEABILITY MATRIX

See Appendix B.

## 4. ACKNOWLEDGMENTS

This research was supported by the US Department of Energy and the Nuclear Energy Advanced Modeling and Simulation Program.

## REFERENCES

[1] Quality Assurance Requirements for Nuclear Facility Applications. Standard ASME-NQA-1-2008 with NQA-1a-2009 Addenda, American Society of Mechanical Engineers, Three Park Avenue, New York, NY 10016-5990, 2009.

# APPENDIX A. CDASH TEST RESULTS

## APPENDIX A. CDASH TEST RESULTS

The following table was obtained from a CDash report. CDash is a web application included with CMake and CTest. CMake is the VERA build system, and CTest is a program that automatically runs the tests defined with CMake and sends results to CDash. All tests are included in the Git repositories as noted in List 1. Appendix B provides a list of the test names, along with the test input files and requirements.

**Table 1. VERAIn Test Report from CDash**

| No. | Test Name | Details | Status |
|---|---|---|---|
| 1 | VERAIn_react2xml_bwr-peach-2 | Completed | Passed |
| 2 | VERAIn_react2xml_bwr-peach-1 | Completed | Passed |
| 3 | VERAIn_react2xml_bwr-peach-4 | Completed | Passed |
| 4 | VERAIn_react2xml_bwr-ge9 | Completed | Passed |
| 5 | VERAIn_react2xml_bwr-peach-5 | Completed | Passed |
| 6 | VERAIn_react2xml_bwr-peach-3 | Completed | Passed |
| 7 | VERAIn_react2xml_bwr-thickthincb | Completed | Passed |
| 8 | VERAIn_react2xml_bwr-peach-6 | Completed | Passed |
| 9 | VERAIn_react2xml_bwr-plr | Completed | Passed |
| 10 | VERAIn_react2xml_bwr-azixs | Completed | Passed |
| 11 | VERAIn_react2xml_bwr-hetblade | Completed | Passed |
| 12 | VERAIn_react2xml_bwr-p6-test | Completed | Passed |
| 13 | VERAIn_react2xml_bwr_jagged | Completed | Passed |
| 14 | VERAIn_react2xml_bwr-axialvoid | Completed | Passed |
| 15 | VERAIn_react2xml_ex7_bwr | Completed | Passed |
| 16 | VERAIn_react2xml_apr1400-cyc1 | Completed | Passed |
| 17 | VERAIn_react2xml_FAST_input_block | Completed | Passed |
| 18 | VERAIn_react2xml_chen_3x3 | Completed | Passed |
| 19 | VERAIn_react2xml_cross_w_baffle | Completed | Passed |
| 20 | VERAIn_react2xml_MSED | Completed | Passed |
| 21 | VERAIn_react2xml_default_density_change | Completed | Passed |
| 22 | VERAIn_react2xml_fast_3x3rod_HFP | Completed | Passed |
| 23 | VERAIn_react2xml_p6a_sa_1300_vuq | Completed | Passed |
| 24 | VERAIn_react2xml_mspc | Completed | Passed |
| 25 | VERAIn_react2xml_fuel_options | Completed | Passed |
| 26 | VERAIn_react2xml_ctf_convergence_terms | Completed | Passed |
| 27 | VERAIn_react2xml_mox1 | Completed | Passed |
| 28 | VERAIn_react2xml_mamba_mass_bal | Completed | Passed |
| 29 | VERAIn_react2xml_detector_edits | Completed | Passed |
| 30 | VERAIn_react2xml_par_cross | Completed | Passed |
| 31 | VERAIn_react2xml_misc_options | Completed | Passed |
| 32 | VERAIn_react2xml_pwr-CE-2a | Completed | Passed |
| 33 | VERAIn_react2xml_pwr_CE_single_16x16 | Completed | Passed |
| 34 | VERAIn_react2xml_pwr-CE-2b | Completed | Passed |
| 35 | VERAIn_react2xml_Sample_3x3 | Completed | Passed |
| 36 | VERAIn_react2xml_Sample_7x7 | Completed | Passed |
| 37 | VERAIn_react2xml_Sample_3x3_multistate | Completed | Passed |
| 38 | VERAIn_react2xml_Sample_full_core_maps | Completed | Passed |

| No. | Test Name | Details | Status |
|---|---|---|---|
| 39 | VERAIn_react2xml_Sample_full_core | Completed | Passed |
| 40 | VERAIn_react2xml_Sample_full_core_matmix | Completed | Passed |
| 41 | VERAIn_react2xml_Sample_full_core_wsym_assm_even | Completed | Passed |
| 42 | VERAIn_react2xml_Sample_full_core_fuelmix | Completed | Passed |
| 43 | VERAIn_react2xml_Sample_full_core_reflector | Completed | Passed |
| 44 | VERAIn_react2xml_Sample_full_core_wsym | Completed | Passed |
| 45 | VERAIn_react2xml_Sample_full_core_wsym_even | Completed | Passed |
| 46 | VERAIn_react2xml_Sample_full_core_wsym_state_assm | Completed | Passed |
| 47 | VERAIn_react2xml_Sample_full_core_wsym_state_dsym | Completed | Passed |
| 48 | VERAIn_react2xml_Sample_full_core_wsym_state_assm_matopt | Completed | Passed |
| 49 | VERAIn_react2xml_Sample_p3 | Completed | Passed |
| 50 | VERAIn_react2xml_Sample_p3_Anderson | Completed | Passed |
| 51 | VERAIn_react2xml_Sample_p3_mpact | Completed | Passed |
| 52 | VERAIn_react2xml_Sample_p3_unit_test | Completed | Passed |
| 53 | VERAIn_react2xml_Sample_p4a_unit_test | Completed | Passed |
| 54 | VERAIn_react2xml_Sample_full_core_wsym_state | Completed | Passed |
| 55 | VERAIn_react2xml_Sample_p4 | Completed | Passed |
| 56 | VERAIn_react2xml_Sample_p6a | Completed | Passed |
| 57 | VERAIn_react2xml_Sample_p6a_multifuel | Completed | Passed |
| 58 | VERAIn_react2xml_Sample_p4_mpact | Completed | Passed |
| 59 | VERAIn_react2xml_Sample_p7 | Completed | Passed |
| 60 | VERAIn_react2xml_Seabrook_5x5rod_HFP | Completed | Passed |
| 61 | VERAIn_react2xml_Sample_qtr_core_wsym_3x3_standalone | Completed | Passed |
| 62 | VERAIn_react2xml_Sample_p9_shuffle | Completed | Passed |
| 63 | VERAIn_react2xml_Sample_p9_restart | Completed | Passed |
| 64 | VERAIn_react2xml_Sample_p9_jumpin | Completed | Passed |
| 65 | VERAIn_react2xml_small_3x3rod_Bison | Completed | Passed |
| 66 | VERAIn_react2xml_small_3x3rod_HFP_2fuel | Completed | Passed |
| 67 | VERAIn_react2xml_small_3x3rod_Bison_standalone | Completed | Passed |
| 68 | VERAIn_react2xml_small_3x3rod_BProd | Completed | Passed |
| 69 | VERAIn_react2xml_small_3x3rod_HFP | Completed | Passed |
| 70 | VERAIn_react2xml_small_3x3rod_HFP_8g | Completed | Passed |
| 71 | VERAIn_react2xml_small_3x3rod_HZP | Completed | Passed |
| 72 | VERAIn_react2xml_small_5cross_assem_3x3rod_HFP | Completed | Passed |
| 73 | VERAIn_react2xml_small_multibundle | Completed | Passed |
| 74 | VERAIn_react2xml_small_3x3rod_IFBA | Completed | Passed |
| 75 | VERAIn_react2xml_small_qtr_core_3x3rod_HFP | Completed | Passed |
| 76 | VERAIn_react2xml_small_multibundle_par | Completed | Passed |
| 77 | VERAIn_react2xml_small_singlerod_HFP_8gb | Completed | Passed |
| 78 | VERAIn_react2xml_small_singlerod_HFP_8g | Completed | Passed |
| 79 | VERAIn_react2xml_small_singlerod_HFP | Completed | Passed |
| 80 | VERAIn_react2xml_5cross_nodal_par_qtr | Completed | Passed |
| 81 | VERAIn_react2xml_decreasing-axial | Completed | Passed |
| 82 | VERAIn_react2xml_front-dash | Completed | Passed |
| 83 | VERAIn_react2xml_disable_xml2ctf | Completed | Passed |

| No. | Test Name | Details | Status |
|---|---|---|---|
| 84 | VERAIn_veradiff_match | Completed | Passed |
| 85 | VERAIn_veradiff_fail | Completed | Passed |
| 86 | VERAIn_react2xml_test_cobra_edits | Completed | Passed |
| 87 | VERAIn_react2xml_test_cobra_edits_inimat | Completed | Passed |
| 88 | VERAIn_react2xml_test_verashift_3a | Completed | Passed |
| 89 | VERAIn_react2xml_test-short | Completed | Passed |
| 90 | VERAIn_react2xml_test_verashift_3a_forward | Completed | Passed |
| 91 | VERAIn_react2xml_smr-7x7 | Completed | Passed |
| 92 | VERAIn_react2xml_test_verashift_3a_cadis | Completed | Passed |
| 93 | VERAIn_react2xml_with-dash | Completed | Passed |
| 94 | VERAIn_react2xml_test_verashift_4mini_nonunique | Completed | Passed |
| 95 | VERAIn_react2xml_2a_LS | Completed | Passed |
| 96 | VERAIn_react2xml_2a_LS_default | Completed | Passed |
| 97 | VERAIn_react2xml_5cross_multiassem | Completed | Passed |
| 98 | VERAIn_react2xml_multi_scalar_error | Completed | Passed |
| 99 | VERAIn_react2xml_test_verashift_small_core_bioshield_no_pv | Completed | Passed |
| 100 | VERAIn_react2xml_test_verashift_4mini_nonunique_bioshield | Completed | Passed |
| 101 | VERAIn_xmlthexp_Sample_p6a | Completed | Passed |
| 102 | VERAIn_react2xml_test_verashift_small_core_bioshield | Completed | Passed |
| 103 | VERAIn_testVERAInMan | Completed | Passed |

# APPENDIX B. REQUIREMENTS AND TEST TRACEABILITY MATRIX

## APPENDIX B. REQUIREMENTS AND TEST TRACEABILITY MATRIX

For entries in the *Additional Info* field, the #NNNN is the VERA ticket number hyperlinked to the appropriate webpage.

**Table 2. Requirements**

| Req. ID | *Requirement Description*<br>*Test Name*<br>*Test Input*<br>*Additional Info* |
|---|---|
| 1 | The VERAIn package shall be able to parse VERA Progression Problem 2A with specified linear source options<br>VERAIn_react2xml_2a_LS<br>2a_LS.inp<br>None |
| 2 | The VERAIn package shall be able to parse VERA Progression Problem 2A with default linear source options<br>VERAIn_react2xml_2a_LS_default<br>2a_LS_default.inp<br>None |
| 3 | React2xml shall support definition of a channel-dependent form loss coefficient map for each spacer grid in the model<br>VERAIn_react2xml_5cross_multiassem<br>5cross_multiassem.inp<br># 3547 |
| 4 | React2xml shall allow spacer grid channel-dependent form loss coefficient maps to be defined using octant symmetry<br>VERAIn_react2xml_5cross_multiassem<br>5cross_multiassem.inp<br>None |
| 5 | React2xml shall allow the user to set modeling coefficients in the Yao-Hochreiter-Leech model for each spacer grid definition<br>VERAIn_react2xml_5cross_multiassem<br>5cross_multiassem.inp<br>None |
| 6 | VERAIO shall support the user inputting a domain decomposition map that can be used by CTF for decomposing the model for parallel solution<br>VERAIn_react2xml_5cross_nodal_par_qtr<br>5cross_nodal_par_qtr.inp<br>None |
| 7 | The VERAIn package shall be able to parse all FAST input options correctly<br><br>FAST_input_block.inp<br>None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 8 | The VERAIn package shall be able to parse inputs in which multigrid solver options are specified |
| | VERAIn_react2xml_MSED |
| | MSED.inp |
| | None |
| 9 | The VERAIn package shall be able to parse a full 3D core with an even core size |
| | |
| | S_even.inp |
| | None |
| 10 | The VERAIn package shall be able to parse 9 3D 17x17 assemblies with feedback enabled in full symmetry |
| | VERAIn_react2xml_Sample_3x3 |
| | Sample_3x3.inp |
| | None |
| 11 | The VERAIn package shall be able to parse multiple state blocks containing differing options selected for the tinlet card and table input for inlet temperature as a function of power |
| | VERAIn_react2xml_Sample_3x3_multistate |
| | Sample_3x3_multistate.inp |
| | None |
| 12 | The VERAIn package shall be able to parse 49 3D 17x17 assemblies with feedback enabled in full symmetry |
| | VERAIn_react2xml_Sample_7x7 |
| | Sample_7x7.inp |
| | None |
| 13 | The VERAIn package shall be able to parse a full 3D core |
| | VERAIn_react2xml_Sample_full_core |
| | Sample_full_core.inp |
| | None |
| 14 | The VERAIn package shall be able to parse a full 3D core with fuel mixing |
| | VERAIn_react2xml_Sample_full_core_fuelmix |
| | Sample_full_core_fuelmix.inp |
| | None |
| 15 | The VERAIn package shall be able to parse a full 3D core with a combination of full, quarter, and octant symmetric maps |
| | VERAIn_react2xml_Sample_full_core_maps |
| | Sample_full_core_maps.inp |
| | None |
| 16 | The VERAIn package shall be able to parse a full 3D core with material mixing |
| | VERAIn_react2xml_Sample_full_core_matmix |
| | Sample_full_core_matmix.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 17 | The VERAIn package shall be able to parse a full 3D core with upper and lower reflector regions |
| | VERAIn_react2xml_Sample_full_core_reflector |
| | Sample_full_core_reflector.inp |
| | None |
| 18 | The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters enabled |
| | VERAIn_react2xml_Sample_full_core_wsym |
| | Sample_full_core_wsym.inp |
| | None |
| 19 | The VERAIn package shall be able to parse a full 3D core with 18x18 assemblies |
| | VERAIn_react2xml_Sample_full_core_wsym_assm_even |
| | Sample_full_core_wsym_assm_even.inp |
| | None |
| 20 | The VERAIn package shall be able to parse a full 3D core of size 16 with numerous BISON parameters enabled |
| | VERAIn_react2xml_Sample_full_core_wsym_even |
| | Sample_full_core_wsym_even.inp |
| | None |
| 21 | The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters enabled with multiple states |
| | VERAIn_react2xml_Sample_full_core_wsym_state |
| | Sample_full_core_wsym_state.inp |
| | None |
| 22 | The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters enabled with multiple states where the assemblies are defined over multiple ASSEMBLY blocks rather than all being defined in the same ASSEMBLY block |
| | VERAIn_react2xml_Sample_full_core_wsym_state_assm |
| | Sample_full_core_wsym_state_assm.inp |
| | None |
| 23 | The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters, including material thermal expansion options |
| | VERAIn_react2xml_Sample_full_core_wsym_state_assm_matopt |
| | Sample_full_core_wsym_state_assm_matopt.inp |
| | None |
| 24 | The VERAIn package shall be able to parse a full 3D core with numerous BISON parameters enabled with multiple states in which one of the lattice definitions is twice as large as the others |
| | VERAIn_react2xml_Sample_full_core_wsym_state_dsym |
| | Sample_full_core_wsym_state_dsym.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 25 | The VERAIn package shall be able to parse Problem 3, which is a single 3D 17x17 assembly at BOL without feedback |
| | VERAIn_react2xml_Sample_p3 |
| | Sample_p3.inp |
| | None |
| 26 | The VERAIn package shall be able to parse nonlinear Anderson Acceleration input options |
| | VERAIn_react2xml_Sample_p3_Anderson |
| | Sample_p3_Anderson.inp |
| | None |
| 27 | The VERAIn package shall be able to parse Problem 3, which is a single 3D 17x17 assembly at BOL without feedback with numerous MPACT options |
| | VERAIn_react2xml_Sample_p3_mpact |
| | Sample_p3_mpact.inp |
| | None |
| 28 | The VERAIn package shall be able to parse Problem 3, which is a single 3D 17x17 assembly at BOL without feedback, where the materials are defined using an include statement |
| | VERAIn_react2xml_Sample_p3_unit_test |
| | Sample_p3_unit_test.inp |
| | None |
| 29 | The VERAIn package shall be able to parse Problem 4, which comprises 9 3D 17x17 assemblies at BOL without feedback |
| | VERAIn_react2xml_Sample_p4 |
| | Sample_p4.inp |
| | None |
| 30 | The VERAIn package shall be able to parse Problem 4, which comprises 9 3D 17x17 assemblies at BOL without feedback with numerous MPACT options |
| | VERAIn_react2xml_Sample_p4_mpact |
| | Sample_p4_mpact.inp |
| | None |
| 31 | The VERAIn package shall be able to parse Problem 4a, which is a single 3D 17x17 assembly with a control rod at BOL without feedback |
| | VERAIn_react2xml_Sample_p4a_unit_test |
| | Sample_p4a_unit_test.inp |
| | None |
| 32 | The VERAIn package shall be able to parse Problem 6a, which is a single 3D 17x17 assembly at BOL with feedback |
| | VERAIn_react2xml_Sample_p6a |
| | Sample_p6a.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 33 | The VERAIn package shall be able to parse a modified Problem 6a, which is a single 3D 17x17 assembly at BOL with feedback with more complicated fuel types |
| | VERAIn_react2xml_Sample_p6a_multifuel |
| | Sample_p6a_multifuel.inp |
| | # 757 |
| 34 | The VERAIn package shall be able to parse Problem 7, which is a full core 3D problem with feedback |
| | VERAIn_react2xml_Sample_p7 |
| | Sample_p7.inp |
| | None |
| 35 | The VERAIn package shall be able to parse Problem 9, which is a full core 3D problem with feedback and depletion, using the jumpin capability |
| | VERAIn_react2xml_Sample_p9_jumpin |
| | Sample_p9_jumpin.inp |
| | None |
| 36 | The VERAIn package shall be able to parse Problem 9, which is a full core 3D problem with feedback and depletion, using the restart capability |
| | VERAIn_react2xml_Sample_p9_restart |
| | Sample_p9_restart.inp |
| | None |
| 37 | The VERAIn package shall be able to parse Problem 9, which is a full core 3D problem with feedback and depletion, using the fuel shuffle capability |
| | VERAIn_react2xml_Sample_p9_shuffle |
| | Sample_p9_shuffle.inp |
| | None |
| 38 | The VERAIn package shall be able to parse a 3D core of 3x3 assemblies |
| | VERAIn_react2xml_Sample_qtr_core_wsym_3x3_standalone |
| | Sample_qtr_core_wsym_3x3_standalone.inp |
| | None |
| 39 | The VERAIn package shall be able to parse a 3D Seabrook 5x5 assembly to test DNB runs |
| | VERAIn_react2xml_Seabrook_5x5rod_HFP |
| | Seabrook_5x5rod_HFP.inp |
| | None |
| 40 | The VERAIn package shall be able to parse a 2D APR1000 Initial core in quarter symmetry |
| | VERAIn_react2xml_apr1400-cyc1 |
| | apr1400-cyc1.inp |
| | None |
| 41 | The VERAIn package shall be able to parse inputs in which a BWR axial_void distribution is specified |
| | VERAIn_react2xml_bwr-axialvoid |
| | bwr-axialvoid.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 42 | The VERAIn package shall be able to process the azimuthal XS option |
| | VERAIn_react2xml_bwr-azixs |
| | bwr-azixs.inp |
| | # 3806 |
| 43 | The VERAIn package shall be able to parse a 2D GE9 BWR lattice |
| | VERAIn_react2xml_bwr-ge9 |
| | bwr-ge9.inp |
| | None |
| 44 | VERAIn shall accept specification of the two-phase friction model which CTF shall use |
| | VERAIn_react2xml_bwr-p6-test |
| | bwr-p6-test.inp |
| | None |
| 45 | VERAIn shall accept specification of the flow regime map to use in CTF |
| | VERAIn_react2xml_bwr-p6-test |
| | bwr-p6-test.inp |
| | None |
| 46 | VERAIn shall accept specification of the interfacial drag model to use in CTF |
| | VERAIn_react2xml_bwr-p6-test |
| | bwr-p6-test.inp |
| | None |
| 47 | The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 7x7 assembly with an 80 mil channel in full symmetry with no gad and no water rods |
| | VERAIn_react2xml_bwr-peach-1 |
| | bwr-peach-1.inp |
| | None |
| 48 | The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 7x7 assembly with an 80 mil channel in full symmetry with 3% gad pins and no water rods |
| | VERAIn_react2xml_bwr-peach-2 |
| | bwr-peach-2.inp |
| | None |
| 49 | The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 7x7 assembly with an 80 mil channel in full symmetry with 3 and 4% gad pins and no water rods |
| | VERAIn_react2xml_bwr-peach-3 |
| | bwr-peach-3.inp |
| | None |
| 50 | The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 8x8 assembly with a 100 mil channel in full symmetry with 3% gad pins and large water rods |
| | VERAIn_react2xml_bwr-peach-4 |
| | bwr-peach-4.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 51 | The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 8x8 assembly with a 100 mil channel in full symmetry with 2% gad pins and large water rods |
| | VERAIn_react2xml_bwr-peach-5 |
| | bwr-peach-5.inp |
| | None |
| 52 | VERAIO shall support setting the tolerance for the BWR pressure balance iteration loop that is performed by CTF |
| | VERAIn_react2xml_bwr-peach-6 |
| | bwr-peach-6.inp |
| | None |
| 53 | The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 8x8 assembly with a 100 mil channel in full symmetry with 2% gad pins and two large water rods |
| | VERAIn_react2xml_bwr-peach-6 |
| | bwr-peach-6.inp |
| | None |
| 54 | The VERAIn package shall be able to parse the default material name 'cool' |
| | VERAIn_react2xml_bwr-plr |
| | bwr-plr.inp |
| | # 4306 |
| 55 | The VERAIn package shall be able to parse a 2D BWR Peach Bottom 2 8x8 assembly with 100 mil corners and 80 mil channels in full symmetry with 2% gad pins and large water rods |
| | VERAIn_react2xml_bwr-thickthincb |
| | bwr-thickthincb.inp |
| | None |
| 56 | VERAIn shall allow the user to specify an orifice map for BWR models which allows for the definition of inlet form losses and flow areas |
| | VERAIn_react2xml_bwr_jagged |
| | bwr_jagged.inp |
| | None |
| 57 | The VERAIn package shall be able to parse a single 3D 3x3 assembly with TH feedback enabled |
| | VERAIn_react2xml_chen_3x3 |
| | chen_3x3.inp |
| | None |
| 58 | The VERAIn package shall be able to parse five 3D 3x3 assemblies with TH feedback enabled arranged in a cross with a baffle |
| | VERAIn_react2xml_cross_w_baffle |
| | cross_w_baffle.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 59 | The VERAIn package shall be able to parse a single 3D 17x17 assembly with TH feedback enabled and CTF convergence criteria specified |
| | VERAIn_react2xml_ctf_convergence_terms |
| | ctf_convergence_terms.inp |
| | None |
| 60 | The VERAIn package shall throw an error for an input in which the axial heights are listed in descending order; the parser shall appropriately fail and describe input error |
| | VERAIn_react2xml_decreasing-axial |
| | decreasing-axial.inp |
| | None |
| 61 | The VERAIn package shall be able to parse inputs in which the default density has been changed |
| | VERAIn_react2xml_default_density_change |
| | default_density_change.inp |
| | None |
| 62 | VERAIn shall properly parse input cards for a user-specified detector edits mesh |
| | VERAIn_react2xml_detector_edits |
| | detector_edits.inp |
| | # 3620 |
| 63 | The VERAIn package shall be able to parse inputs in which xml2ctf is disabled during VERA init |
| | VERAIn_react2xml_disable_xml2ctf |
| | disable_xml2ctf.inp |
| | None |
| 64 | VERAIn shall process BWR blade and detector maps |
| | VERAIn_react2xml_ex7_bwr |
| | ex7_bwr.inp |
| | None |
| 65 | The VERAIn package shall be able to parse a single 3D 3x3 assembly at hot full power |
| | VERAIn_react2xml_fast_3x3rod_HFP |
| | fast_3x3rod_HFP.inp |
| | None |
| 66 | The VERAIn package shall throw an error for an input in which a hyphen is given before the assembly name in the assembly map. The parser shall appropriately fail and describe the input error |
| | VERAIn_react2xml_front-dash |
| | front-dash.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 67 | The VERAIn package shall be able to parse inputs with complex fuel and material specifications |
| | VERAIn_react2xml_fuel_options |
| | fuel_options.inp |
| | None |
| 68 | The VERAIn package shall be able to parse inputs in which the COBRATF/crud and mass balance/steam generator cards are specified |
| | VERAIn_react2xml_mamba_mass_bal |
| | mamba_mass_bal.inp |
| | None |
| 69 | The VERAIn package shall be able to parse inputs in which numerous miscellaneous options are enabled simultaneously |
| | VERAIn_react2xml_misc_options |
| | misc_options.inp |
| | None |
| 70 | The VERAIn package shall be able to parse a single MOX PWR assembly with three different plutonium enrichment zones |
| | VERAIn_react2xml_mox1 |
| | mox1.inp |
| | # 2226 |
| 71 | The VERAIn package shall be able to parse inputs in which multigrid solver options are specified |
| | VERAIn_react2xml_mspc |
| | mspc.inp |
| | None |
| 72 | The VERAIn package shall not be able to parse an input in which multiple scalar parameters are specified on the same line. The parser shall appropriately fail and describe the input error |
| | VERAIn_react2xml_multi_scalar_error |
| | multi_scalar_error.inp |
| | None |
| 73 | The VERAIn package shall be able to parse a single 3D 17x17 assembly with feedback enabled with 1,300 ppm boron |
| | VERAIn_react2xml_p6a_sa_1300_vuq |
| | p6a_sa_1300_vuq.inp |
| | None |
| 74 | The VERAIn package shall be able to parse 5 3D 17x17 assemblies with feedback enabled arranged in a cross with parallel 1 set in the COBRATF block. |
| | VERAIn_react2xml_par_cross |
| | par_cross.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 75 | The VERAIn package shall be able to parse a single 16x16 CE design PWR assembly with Erbia and two enrichments |
| | VERAIn_react2xml_pwr-CE-2a |
| | pwr-CE-2a.inp |
| | None |
| 76 | The VERAIn package shall be able to parse a single 16x16 CE design PWR assembly with Erbia and three enrichments |
| | VERAIn_react2xml_pwr-CE-2b |
| | pwr-CE-2b.inp |
| | None |
| 77 | The VERAIn package shall be able to parse a single 16x16 CE design PWR assembly |
| | VERAIn_react2xml_pwr_CE_single_16x16 |
| | pwr_CE_single_16x16.inp |
| | None |
| 78 | The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with a natural blanket |
| | VERAIn_react2xml_small_3x3rod_BProd |
| | small_3x3rod_BProd.inp |
| | None |
| 79 | The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with numerous BISON options enabled. |
| | VERAIn_react2xml_small_3x3rod_Bison |
| | small_3x3rod_Bison.inp |
| | None |
| 80 | The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with numerous BISON options enabled with solve_type=standalone |
| | VERAIn_react2xml_small_3x3rod_Bison_standalone |
| | small_3x3rod_Bison_standalone.inp |
| | None |
| 81 | The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP |
| | VERAIn_react2xml_small_3x3rod_HFP |
| | small_3x3rod_HFP.inp |
| | None |
| 82 | The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with two fuel enrichments |
| | VERAIn_react2xml_small_3x3rod_HFP_2fuel |
| | small_3x3rod_HFP_2fuel.inp |
| | None |
| 83 | The VERAIn package shall be able to parse a single 3D 3x3 assembly at 10% power |
| | VERAIn_react2xml_small_3x3rod_HFP_8g |
| | small_3x3rod_HFP_8g.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 84 | The VERAIn package shall allow the user to set the solve_heat_end option in the COBRATF input block |
| | VERAIn_react2xml_small_3x3rod_HFP_8g |
| | small_3x3rod_HFP_8g.inp |
| | None |
| 85 | The VERAIn package shall be able to parse a single 3D 3x3 assembly at hot zero power |
| | VERAIn_react2xml_small_3x3rod_HZP |
| | small_3x3rod_HZP.inp |
| | None |
| 86 | The VERAIn package shall be able to parse a single 3D 3x3 assembly at HFP with IFBA pins |
| | VERAIn_react2xml_small_3x3rod_IFBA |
| | small_3x3rod_IFBA.inp |
| | None |
| 87 | The VERAIn package shall be able to parse a five 3D 3x3 assemblies at HFP when arranged in a cross |
| | VERAIn_react2xml_small_5cross_assem_3x3rod_HFP |
| | small_5cross_assem_3x3rod_HFP.inp |
| | None |
| 88 | The VERAIn package shall be able to parse a five 3D 3x3 assemblies at 10% power when arranged in a cross with numerous COUPLING block options |
| | VERAIn_react2xml_small_multibundle |
| | small_multibundle.inp |
| | None |
| 89 | The VERAIn package shall be able to parse a five 3D 3x3 assemblies at 10% power when arranged in a cross in quarter symmetry with numerous COUPLING block options and when CTF is run in parallel |
| | VERAIn_react2xml_small_multibundle_par |
| | small_multibundle_par.inp |
| | None |
| 90 | The VERAIn package shall be able to parse a full core of 3D 3x3 assemblies at HFP in quarter symmetry |
| | VERAIn_react2xml_small_qtr_core_3x3rod_HFP |
| | small_qtr_core_3x3rod_HFP.inp |
| | None |
| 91 | The VERAIn package shall be able to parse a single 3D fuel pin at HFP in full symmetry |
| | VERAIn_react2xml_small_singlerod_HFP |
| | small_singlerod_HFP.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 92 | The VERAIn package shall be able to parse a single 3D fuel pin at HFP in full symmetry with materials defined in an include file |
| | VERAIn_react2xml_small_singlerod_HFP_8g |
| | small_singlerod_HFP_8g.inp |
| | None |
| 93 | The VERAIn package shall be able to parse a single 3D fuel pin at HFP in full symmetry with materials defined in an include file with a boron search enabled |
| | VERAIn_react2xml_small_singlerod_HFP_8gb |
| | small_singlerod_HFP_8gb.inp |
| | None |
| 94 | The VERAIn package shall be able to parse a full 3D SMR core in quarter symmetry with feedback enabled |
| | VERAIn_react2xml_smr-7x7 |
| | smr-7x7.inp |
| | None |
| 95 | The VERAIn package shall be able to parse a single 3D fuel pin at HFP in full symmetry with a shortened height |
| | VERAIn_react2xml_test-short |
| | test-short.inp |
| | None |
| 96 | The VERAIn package shall be able to parse a 3D 3x3 assembly at HFP in full symmetry with CTF edits enabled |
| | VERAIn_react2xml_test_cobra_edits |
| | test_cobra_edits.inp |
| | None |
| 97 | The VERAIn package shall be able to parse a 3D 3x3 assembly at HFP in full symmetry with CTF edits enabled when material definitions are increased by 100 |
| | VERAIn_react2xml_test_cobra_edits_inimat |
| | test_cobra_edits_inimat.inp |
| | None |
| 98 | The VERAIn package shall be able to parse a 3D 17x17 assembly with no feedback in quarter symmetry and with VERASHIFT options enabled |
| | VERAIn_react2xml_test_verashift_3a |
| | test_verashift_3a.inp |
| | None |
| 99 | The VERAIn package shall be able to parse a 3D 17x17 assembly with no feedback in quarter symmetry and with VERASHIFT CADIS options enabled |
| | VERAIn_react2xml_test_verashift_3a_cadis |
| | test_verashift_3a_cadis.inp |
| | None |

| Req. ID | *Requirement Description* |
|---|---|
| | *Test Name* |
| | *Test Input* |
| | *Additional Info* |
| 100 | The VERAIn package shall be able to parse a 3D 17x17 assembly with no feedback in quarter symmetry and with VERASHIFT problem_mode=forward enabled |
| | VERAIn_react2xml_test_verashift_3a_forward |
| | test_verashift_3a_forward.inp |
| | None |
| 101 | The VERAIn package shall be able to parse nine 3D 17x17 assemblies with no feedback in quarter symmetry and with VERASHIFT options enabled |
| | VERAIn_react2xml_test_verashift_4mini_nonunique |
| | test_verashift_4mini_nonunique.inp |
| | None |
| 102 | The VERAIn package shall throw an error for an input modeled in quarter symmetry with VERAShift options enabled and with detectors and a bioshield defined; the vera2omn script shall produce the following— Error: core_translate parameter should match outer vessel radius. Writing Omnibus excore input file Error: detector cannot extend into vessel VERA2OMN failed |
| | VERAIn_react2xml_test_verashift_4mini_nonunique_bioshield |
| | test_verashift_4mini_nonunique_bioshield.inp |
| | None |
| 103 | The VERAIn package shall be able to parse a 5x5 mini core with no feedback in quarter symmetry and with automated ex-core detector and bioshield setup with a vessel; the vera2omn script should be triggered and should produce the proper ex-core geometry input file |
| | VERAIn_react2xml_test_verashift_small_core_bioshield |
| | test_verashift_small_core_bioshield.inp |
| | None |
| 104 | The VERAIn package shall throw an error for an input modeled in quarter symmetry and with automated ex-core detector and bioshield without a vessel definition; the vera2omn script should be triggered, and the following error message produced— Error: vera_pressure_vessel is False but automated excore generator will only define the geometry beyond the vessel. VERA2OMN failed |
| | VERAIn_react2xml_test_verashift_small_core_bioshield_no_pv |
| | test_verashift_small_core_bioshield_no_pv.inp |
| | None |
| 105 | The VERAIn package shall be able to parse inputs that have dashes in the assembly names within the assm_map |
| | VERAIn_react2xml_with-dash |
| | with-dash.inp |
| | None |